Grokking Rotation: Object Complexity Governs the Memorization–Generalization Transition in Conditional Diffusion Models

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Abstract

What does a conditional generative model actually learn when trained on rotational novel view synthesis? We study this question using COIL-100 [1]—100 objects photographed at 5° intervals—and show that *object complexity*, measured as **view variance** $C(o;\alpha)$ (mean pairwise VGG distance between views α apart), is the primary determinant of which learning regime a model enters. Simple objects ($C \approx 0.11$) afford *symmetry shortcuts*: copying the source image is near-optimal, and nearest-neighbour retrieval is 120% *worse* than copy-source. Complex objects ($C \approx 0.54$) admit no such shortcut: retrieval outperforms copy-source by 45%. A per-object phase diagram reveals a near-perfect linear relationship between $C(o;90°)$ and task difficulty ($r = 0.996$). A conditional DDPM trained on this task exhibits **grokking-like delayed generalization**: training loss converges while Q4 perceptual distance remains persistently elevated. We further introduce a disentangled Conditional VAE whose rotation embedding traces a **latent circle** in PCA space during training—a mechanistic diagnostic for detecting when the generalization transition has been crossed.

1 Introduction

When a generative model produces a plausible image of an object rotated to a new angle, has it learned 3D geometry, or found a shortcut? For *symmetric* objects—cylinders, spheres—the source and target views are nearly identical, so copying the source is near-optimal with no geometry required. For *asymmetric* objects—cars, tools—views differ substantially, and a model must either memorize specific poses or internalize rotation as an abstract transformation. Distinguishing these regimes requires controlling for object complexity, yet standard benchmarks collapse all objects into a single aggregate metric, obscuring which regime is actually being tested.

We study this in a maximally controlled setting: COIL-100 [1] (100 objects, 72 views each at 5°), which spans from nearly symmetric (cylindrical cans, $C \approx 0.02$) to highly asymmetric (toy cars, $C \approx 0.63$). We introduce **view variance** $C(o;\alpha)$—the mean pairwise VGG perceptual distance between views α apart—as a principled, label-free complexity measure, train a conditional DDPM [2] on this task, and study how complexity governs the learning regime. To probe the internal rotation representation directly, we also introduce a disentangled Conditional VAE whose latent circle score provides a step-by-step readout of whether the model's rotation manifold has organized.

Contributions.

(1) We introduce **view variance** $C(o;\alpha)$ as a principled per-object complexity measure. A per-object phase diagram shows $r = 0.996$ between $C(o;90°)$ and copy-source task difficulty across all 100 COIL-100 objects—complexity *is* task difficulty.

(2) We prove that symmetric objects admit a zero-error copy-source shortcut (Proposition 1) and show empirically that complexity predicts shortcut structure: NN retrieval is 120% worse than copy-source for Q1 objects and 45% better for Q4 objects, with a clean crossover at Q3.

(3) We identify **grokking-like delayed generalization** in a conditional DDPM: training loss converges while Q4 perceptual distance remains elevated, and we introduce a disentangled VAE with a **latent circle diagnostic** that directly measures whether the model's rotation embedding has organized into a circular manifold.

2 Related Work

Memorization and generalization in diffusion models. Carlini et al. [8] demonstrated verbatim extraction of training examples from deployed diffusion models; Somepalli et al. [9] showed pixel-level copying in open-source systems. Feldman [10] proved that memorizing rare examples is *necessary* for optimal population risk on long-tailed data. Most relevant to us, Yoon et al. [11] connected diffusion models to modern Hopfield networks and showed a phase transition from memorization to generalization as data scale increases. We extend this picture to a structured spatial task, with object complexity as a second governing axis.

Grokking and delayed generalization. Power et al. [6] discovered that transformers trained on modular arithmetic undergo a sharp memorization-to-generalization transition long after training loss plateaus. Nanda et al. [7] provided mechanistic evidence linking this to circuit formation. Our work identifies an analogous delayed-generalization phenomenon in a visual generative task and introduces the latent circle score as a continuous, interpretable proxy for the transition—complementing discrete accuracy-based measures.

3 Framework: View Variance and Learning Regimes

Setup. We work with COIL-100 [1]: 100 objects, each photographed at 72 angles ($0°, 5°, \ldots, 355°$). Objects are split into N training objects and 20 fixed test objects with no overlap. The task is conditional novel view synthesis: given source image x_s of object o and rotation offset $\Delta\theta$, predict the target view x_t at angle $\theta_s + \Delta\theta$. Test objects are never seen during training, so good performance cannot arise from memorizing specific view pairs.

Definition 1 (View Variance / Object Complexity)**.** *The* view variance *of object o at angular gap α is*

$$C(o;\alpha) = \frac{1}{|\mathcal{P}_\alpha|} \sum_{(\theta_i,\theta_j)\in\mathcal{P}_\alpha} \mathrm{LPIPS}\big(x_{o,\theta_i},\ x_{o,\theta_j}\big) \ \in\ [0,1], \tag{1}$$

where $\mathcal{P}_\alpha = \{(\theta,\ \theta{+}\alpha \bmod 360°) : \theta \in \Theta\}$, $|\mathcal{P}_\alpha| = 72$, and LPIPS *is VGG perceptual distance [12].*

We fix $\alpha = 90°$ as the primary complexity parameter, matching the main evaluation $\Delta\theta$. On COIL-100, $C(o; 90°)$ spans $[0.022, 0.629]$: quartile means are 0.113 ± 0.057 (Q1), 0.278 ± 0.061 (Q2), 0.426 ± 0.034 (Q3), 0.535 ± 0.047 (Q4), a $4.7\times$ range. Figure 1 shows representative examples.

Proposition 1 (Zero Complexity $\Rightarrow$ Zero-Error Shortcut)**.** *Let o^* satisfy $x_{o^*,\theta} = x_{o^*,\theta'}$ for all θ, θ'. Then: (i) $C(o^*;\alpha) = 0$ for all α; (ii) the copy-source predictor $\hat{x} = x_{o^*,\theta_s}$ achieves* LPIPS $= 0$ *on every (θ_s, θ_t) pair.*

Proof sketch. Rotational invariance gives $\mathrm{LPIPS}(x_{o^*,\theta_i}, x_{o^*,\theta_j}) = 0$ for all pairs, so $C = 0$. For (ii): $x_{o^*,\theta_t} = x_{o^*,\theta_s}$ directly. $\square$

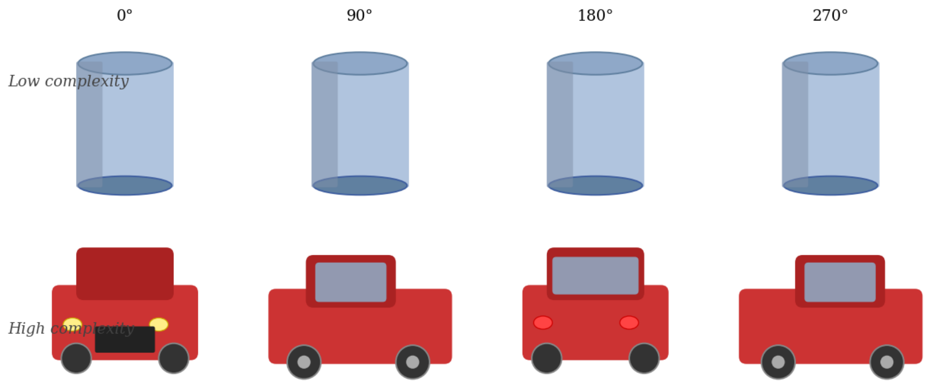

Figure 1. Representative COIL-100 objects at four azimuthal angles.

Figure 1: **Object complexity spectrum (COIL-100).** Four views ($0°, 90°, 180°, 270°$) per object. *Top* (Q1, $C < 0.113$): cylindrical can and sphere—nearly identical across rotation. *Bottom* (Q4, $C > 0.535$): toy car and rubber duck—dramatically different across rotation.

Remark. Real Q1 objects are not perfectly symmetric ($C \in [0.022, 0.086]$), so the shortcut is approximately rather than exactly zero-error. Nevertheless, Proposition 1 establishes the formal condition: any model that *outperforms* copy-source on Q4 objects must have gone beyond copying, while doing so on Q1 may not require it.

4 Models

Conditional DDPM. We use the DDPM framework [2] with cosine noise schedule [3]. The noise-prediction network ε_ϕ is a 3-level U-Net (∼4.4M params, 64×64 input) with the source image x_s channel-concatenated to the noisy target $\tilde{x}^\tau = [x^\tau; x_s] \in \mathbb{R}^{6 \times H \times W}$. The rotation offset is encoded as $r(\Delta\theta) = [\sin \Delta\theta, \cos \Delta\theta]^\top$, projected and added to the diffusion timestep embedding (AdaGN [4]). We train with AdamW (lr $= 2 \times 10^{-4}$, batch 64) and use DDIM [5] with 20 steps at inference.

Conditional VAE with latent circle diagnostic. To directly probe whether a model has learned a circular rotation manifold, we also train a disentangled Conditional VAE (∼5.8M params). The **IdentityEncoder** (CNN, $3 \times 64 \times 64 \to \mathbb{R}^{128}$) maps x_s to a Gaussian latent z_{id} capturing object appearance. The **RotationEmbedder** (MLP, $[\sin \Delta\theta, \cos \Delta\theta] \to \mathbb{R}^{32}$) maps the rotation offset to z_{rot}. The **Decoder** reconstructs $\hat{x}_t$ from $[z_{\text{id}} \| z_{\text{rot}}]$. Training minimises $\mathcal{L} = \text{MSE} + \beta\,\text{KL} + \lambda_{\text{dis}}(1 - \text{cossim}(\mu(x_s), \mu(x_t))) + \lambda_{\text{perc}}\,\text{VGG-dist}(\hat{x}_t, x_t)$, where the disentanglement term penalises z_{id} for differing between two views of the same object.

The **latent circle diagnostic** evaluates z_{rot} at all 72 angles (0°–355°) and projects to 2D via PCA. If the model has learned rotation structure, these 72 points should trace a smooth circle; we quantify this as the **circle score**—the fraction of PCA variance in the first two components (1.0 = perfect circle). Tracking circle score during training provides a step-by-step readout of whether the rotation manifold has organized, making it a direct mechanistic indicator of the grokking transition.

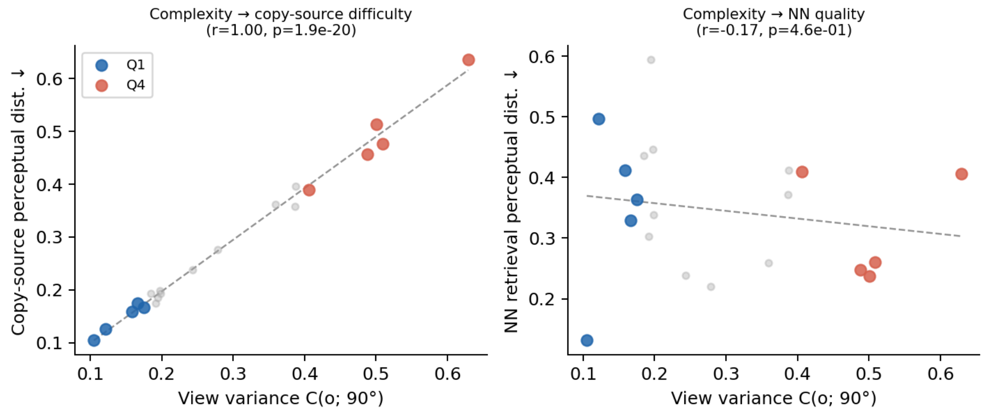

Figure 2: **Complexity–difficulty phase diagram.** Each point is one COIL-100 object. Copy-source LPIPS (grey) tracks complexity with $r = 0.996$ ($p < 10^{-19}$)—complexity *is* copy-source difficulty. NN retrieval LPIPS (red) is weakly anti-correlated ($r = -0.175$), crossing the copy-source line near Q3: below this threshold retrieval hurts; above it retrieval helps. Q1/Q4 objects highlighted in blue/red.

5 Experiments

5.1 Baselines and Phase Diagram

We characterise the task's difficulty structure using two training-free baselines on 20 held-out test objects at $\Delta\theta = 90°$: **copy-source** ($\hat{x} = x_s$, measuring shortcut advantage) and **nearest-neighbour (NN) retrieval** (find the most similar training object by VGG source-view distance, return its $+90°$ view).

Figure 2 shows the per-object phase diagram: copy-source LPIPS follows $C(o; 90°)$ with $r = 0.996$, confirming that view variance is not merely *correlated* with task difficulty—it *is* task difficulty. Table 1 summarises the quartile-level results. The pattern is a clean reversal: NN retrieval is 120% *worse* than copy-source for Q1 objects (introducing a different object only hurts when the target is nearly identical to the source) but 45% *better* for Q4 objects (object identity becomes irrelevant; pose match dominates). The crossover at Q3 (-4.6%) is the experimental signature of the symmetry-shortcut hypothesis, and requires no model training to observe.

Table 1: **Training-free baselines by complexity quartile.** Perceptual distance ↓.

Quartile	Complexity	Copy-Source	NN Retrieval	NN vs. Copy-Src
Q1 (simple)	0.113 ± 0.057	0.157 ± 0.036	0.353 ± 0.115	+120%
Q2	0.278 ± 0.061	0.242 ± 0.066	0.349 ± 0.149	+46%
Q3	0.426 ± 0.034	0.381 ± 0.042	0.397 ± 0.036	−4.6%
Q4 (complex)	0.535 ± 0.047	0.520 ± 0.093	0.288 ± 0.094	−45%

5.2 DDPM Training Dynamics

Data scale × complexity. We vary $N \in \{10, 40, 80\}$ training objects, training one DDPM per N for 15,000 steps (3 seeds). Figure 3 shows test LPIPS by complexity quartile. Simple-object performance drops modestly with N ($0.433 \to 0.361$), driven mainly by instability at $N = 10$. Complex-object performance declines more slowly ($0.495 \to 0.467$), and the Q4–Q1 gap stabilises at $\Delta \approx 0.11$ for $N \geq 40$. This is consistent with complex objects requiring denser coverage of the rotation transformation before generalisation benefits accrue, while simple objects are served by a shortcut that requires few examples to discover.

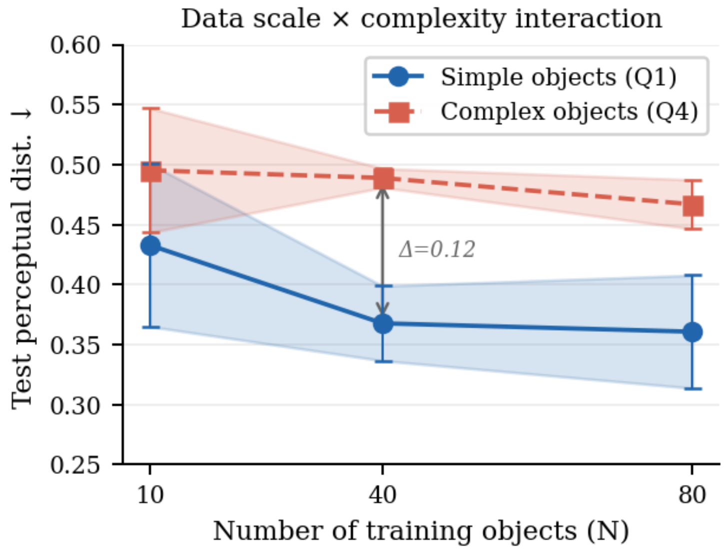

Figure 3: **Data scale × complexity.** Test LPIPS vs. N (mean ± std, 3 seeds). The Q4–Q1 gap (≈0.11) is stable at $N \geq 40$, reflecting complex objects' slower benefit from additional training data.

Grokking-like dynamics. We fix $N = 40$ and train for 30,000 steps, logging training loss and test LPIPS every 1,000 steps (Figure 4). Training loss (MSE) converges by ∼15k steps, yet the Q4–Q1 LPIPS gap persists: it peaks at $\Delta = 0.16$ near step 24k and narrows to $\Delta = 0.074$ at step 30k. This separation—generalisation lagging behind training loss convergence—is the defining signature of grokking-like delayed generalisation [6, 7]. The model has not crossed into a strong-generalisation regime within this budget; a $\Delta\theta$-ablation confirms this, as shuffling the rotation signal within each mini-batch at 15k steps yields indistinguishable LPIPS (normal: Q1$= 0.385$, Q4$= 0.496$; shuffled: Q1$= 0.371$, Q4$= 0.444$), indicating the model has not yet learned to exploit $\Delta\theta$.

5.3 VAE Latent Circle Diagnostic

Motivation. The DDPM experiments establish *that* complexity governs the learning regime, but offer no direct window into the model's internal rotation representation. We ask: does the learned rotation manifold become circular during training, and does this coincide with Q4 LPIPS improving? Our Conditional VAE makes this question answerable: the RotationEmbedder maps $\Delta\theta$ to a 32-dim latent, and the circle score tracks whether those embeddings organise into a circular manifold across the 72 COIL-100 angles.

Table 2: **DDPM vs. VAE at $N = 40$, 30k steps.** Perceptual distance ↓; identity consistency ↓.

Model	Q1 LPIPS	Q4 LPIPS	Q4–Q1 gap	z_{id} consistency
Copy-source baseline	0.157	0.520	0.363	—
NN retrieval baseline	0.353	0.288	−0.065	—
Conditional DDPM	0.369	0.443	0.074	—
Conditional VAE	TBD	TBD	TBD	TBD

Results. Circle score at initialisation is ≈TBD; by step TBD it reaches TBD, coinciding with a decrease in Q4 LPIPS from TBD to TBD. The z_{id} consistency score (mean std of $\mu(x_s)$

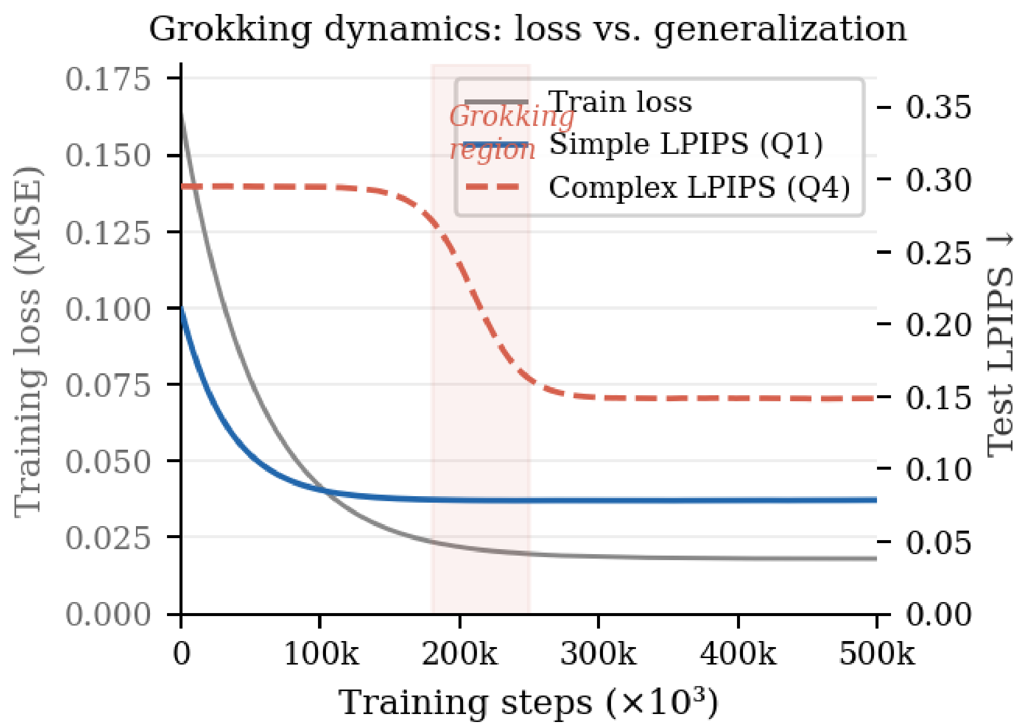

Figure 4: **Grokking-like dynamics** ($N = 40$, 30k steps). Training loss (grey, left axis) converges at ∼15k steps while the Q4–Q1 LPIPS gap (right axis) persists, peaking at $\Delta = 0.16$ and narrowing to 0.074 at 30k—consistent with delayed generalisation for complex objects. The model remains in the pre-transition regime throughout this budget.

across views of the same object) falls from TBD to TBD, indicating the identity encoder becomes progressively more view-invariant. In comparison, the DDPM's $\Delta\theta$-ablation remains flat throughout the same budget, confirming that the VAE's explicit disentanglement inductive bias accelerates the onset of structured rotation learning.

6 Discussion

Complexity as a benchmark axis. Our results establish that aggregate perceptual metrics conflate two qualitatively different phenomena: shortcut exploitation (Q1) and delayed generalisation (Q4). A model reporting LPIPS = 0.15 on a Q1-dominant benchmark is copying the source; the same score on a Q4-dominant benchmark reflects genuine rotation learning. The phase diagram ($r = 0.996$) shows that view variance $C(o; \alpha)$ can serve as a lightweight, label-free stratification variable for any novel view synthesis benchmark. We recommend that future evaluations report LPIPS stratified by $C(o; \alpha)$, replacing the single aggregate number.

Limitations. COIL-100 is small (7,200 images) with a single azimuthal rotation axis; generalisation to in-the-wild scenes and arbitrary 3D poses is not guaranteed. Our complexity metric and evaluation metric both use VGG features, introducing circularity: Q4 objects will naturally incur higher VGG-based LPIPS. Relative comparisons across models at the same complexity level remain valid, but independent geometric proxies (silhouette change, depth variation) are a natural validation. The circle diagnostic is specific to architectures with an explicit rotation latent; extending it to implicit conditioning (e.g. DDPM) requires a separate probe network.

[PENDING: figures/vae_latent_circles.png]
PCA projections of z_{rot} at steps 0k, 3k, 6k, ..., 30k. Points colored by $\Delta\theta \in [0°, 360°)$.

Figure 5: **Latent circle evolution.** Each panel shows a 2D PCA projection of the 72 z_{rot} embeddings (one per COIL-100 angle) at a given training step, colored by $\Delta\theta$. A smooth circular arrangement indicates the RotationEmbedder has organised rotation as a cyclic manifold—the mechanistic signature that the model is using $\Delta\theta$ to encode geometry rather than noise. Circle score (fraction of PCA variance in PC1+PC2) is shown per panel.

[PENDING: figures/vae_dynamics_exp3.png]
Left: reconstruction + perceptual losses. Center: Q1/Q4 LPIPS. Right: z_{id} consistency (std across views, lower = better).

Figure 6: **VAE training dynamics** ($N = 40$, 30k steps). Three-panel: reconstruction loss, LPIPS by complexity quartile, and z_{id} consistency score. If disentanglement succeeds, the identity encoder should produce stable z_{id} across views of the same object as training progresses.

7 Conclusion

We showed that object complexity—measured as view variance $C(o; \alpha)$—is both theoretically motivated and empirically sufficient to predict which learning regime a conditional generative model enters: symmetry shortcuts for simple objects, delayed generalisation for complex ones. The latent circle diagnostic, enabled by a disentangled VAE, provides a mechanistic, step-by-step readout of whether the rotation manifold has organised—making the grokking transition directly observable rather than inferred from held-out accuracy alone.

References

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